

Your cabinet needs carefully manipulated airflow, designed to divert hot air out of the cabinet through its vents, and suck in cool air from outside – the goal is defined air paths, and *not* turbulence. Think of a wind tunnel, with controlled direction of airflow – designing such cabinets is a proper science and not everyone gets it right.

As mentioned don't be fooled by SMPS' power ratings; one decent tip is to note the weight of the unit – the heavier the better. This is because such units will have better quality and larger heatsinks and power capacitor units. With power supplies brands are quite important, more so than cabinets. Modular PSUs are helpful; since you only use the cables you need. This helps prevent cable clutter and optimises airflow. Opt for an SMPS with *Active PFC* (acronym for Active Power Factor Correction). Active PFC uses a dedicated circuit to maintain accurate power distribution. This method of power correction also reduces harmonics (interference) and corrects power delivery according to the AC input voltage. The older method for power correction is called *Passive PFC* and PSUs based on this should be avoided. In case you intend on setting up a gaming rig or have special requirements like SLI or CrossFire support, you need to look at available connections on the PSU. Some of the new graphics cards need eight pin power connects, and for a high-end multi GPU setup you may need up to four such connectors. Obviously power supplies having support for 3-way SLI and quad CrossFire will have matching power outputs to the tune of at least 1 kilowatt. Another noteworthy factor to consider when choosing power supplies is their efficiency which is measured as a percentage. The higher the percentage the higher is the efficiency of the unit. Look for power supplies rated at 80 per cent and above; these are also called 80+ units.

The position of cabinets and power supplies is much better than when I last went shopping for either which was around a year back. By this I refer to availability of bigger brands and certain models. Of course your choice of cabinet and PSU depends on the kind of components in your PC. A gaming rig needs a well ventilated cabinet and a powerful PSU; while an HTPC needs

a really small box-type cabinet and a compact PSU. Casual users may need something simple and affordable.

If you're looking to build an HTPC you shouldn't skimp on the quality of PSU and cabinet. For a case I recommend a really compact ITX form factor. The cube-shaped cabinets are really aesthetic and look very classy and unobtrusive and I prefer them to the slim HTPC cabinets as these are usually quite high and deep. Zebronic has a small ITX cabinet called *Tambi*. It features a handle on the front for easy cartage and space for a single ODD and HDD. There's a small fan provided at the rear for airflow, but this isn't suitable for a powerful HTPC and should be looked at only if you are building a sub-Rs 30,000 HTPC without a graphics card. This is a no-frills cabinet with reasonable build quality but limited expansion and cooling. Antec's *Fusion Remote* is available in black and black/silver. This is a beautiful looking and excellently built cabinet with media management software and remote bundled. It's got space for two HDDs and one ODD. Two adjacent, side mounted 120mm fans provide cooling. The *Fusion Remote* is priced at Rs 10,500 and is a great deal considering the feature set. Equally hot is the cube form factor Antec *NSK1380 EC*. This one has support for three HDDs, one ODD and four full-height expansion cards. Cooling is via a rear 120 mm fan. The *NSK1280 EC* comes with a 350W PSU that's 80+ rated. For Rs 7,500 this is good if you do not need a remote bundle or have your own. CoolerMaster's *Media 260* is another option. It's built well and the body has a lot of venting and includes media management software and a remote. Priced at Rs 8,000 this is a cheaper alternative to Antec's *Fusion Remote* although a PSU isn't bundled, so that will cost more. I recommend the Antec *Fusion Remote*.

For a generic good build I recommend CoolerMaster's 690 – this is a great cabinet that is very well priced at Rs 3,500 and looks superb. The black mesh on the front and top looks classy and doubles as a venting system. With support for five 120mm fans and two bundled the 690 is a wonderful cabinet for any kind of user. If you want something really beautiful, the CM *Cosmos S* is a work of art. With touch controls, all aluminium build, sleek out-of-this-world looks, excellent cooling options that includes a huge 200 mm side fan and a meshed side and front combined with

LED fans this *Cosmos S* is simply lovely. Priced at Rs 13,500 this cabinet demands a pretty penny, but will upgrade the look of your desktop forever. I recommend this for the extreme gamer and/or show-off. Antec's 900 is another terrific option for Rs 8,000. It's much lighter and seems flimsily built for that. A lot of venting and cooling means the 900 handles high-end (read hot) gaming rigs well.

For beefy power supplies look at Corsair's HX and TX series. The HX 620 is a superb option for Rs 6,800 and will power anything other than top-end multi GPU setups (GTX 280s and HD 4870s are too power hungry). Corsair's HX 1000 sits atop the PSU food chain and will handle anything this side of a 3-way GTX 280 plus overclocked quad core setup. Priced at Rs 13,000 the HX 1000 is for the extreme overclocker or gamer with killer rigs. Corsair's TX 650 is available for Rs 6,000 and represents the value entrant to the high-end PSUs. All these are superb options for powerful PCs. Corsair's VX 450 is available for Rs 2,500 and should handle mid-range gaming setups. In fact I recommend this PSU for anyone looking at a decent PC – cheaper PSUs aren't worth the heartache of damaged components later.

CoolerMaster has some good power supplies as well. Their *Real Power* series are noteworthy and should be considered as secondary alternatives to Corsair's units. For someone looking at something between Corsair's HX 620 and HX 1000, CMs *Real Power 850W* may just catch your fancy at Rs 8,500. Antec also has a lot of PSUs in their *Earth Power* series, their *Earth Power 650 watts* is a steal at Rs 6,000. Their *Quattro* series are even better, but too costly for the performance they offer. A new entrant in India, *Tagan* is a known PSU manufacturer and has a few decent options; although they're priced similarly to Corsair and considering the availability of the latter you can skip these entirely.

If I were looking at a top-end setup I'd go with the CM *Cosmos S* and the Corsair HX 1000 – a heavenly abode for your costly components. Given a budget of Rs 10,000 for me it doesn't get better than a CM 690 coupled with a Corsair HX 620. If you must save cash, the VX 450 is a good option for a mid-range PC with the CM 690 remaining my chassis of choice. ☐

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